Università di Genova

MeshBoat

Secure your floating home

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Problem

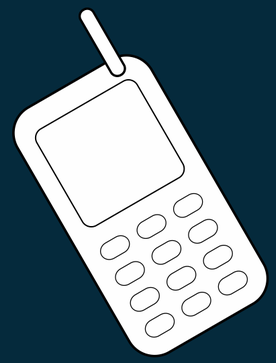
+14%

Maritime crimes compared to the previous year

Looting focusing on high-value, easily transportable assets

131

Maritime crimes in Italy in 2025



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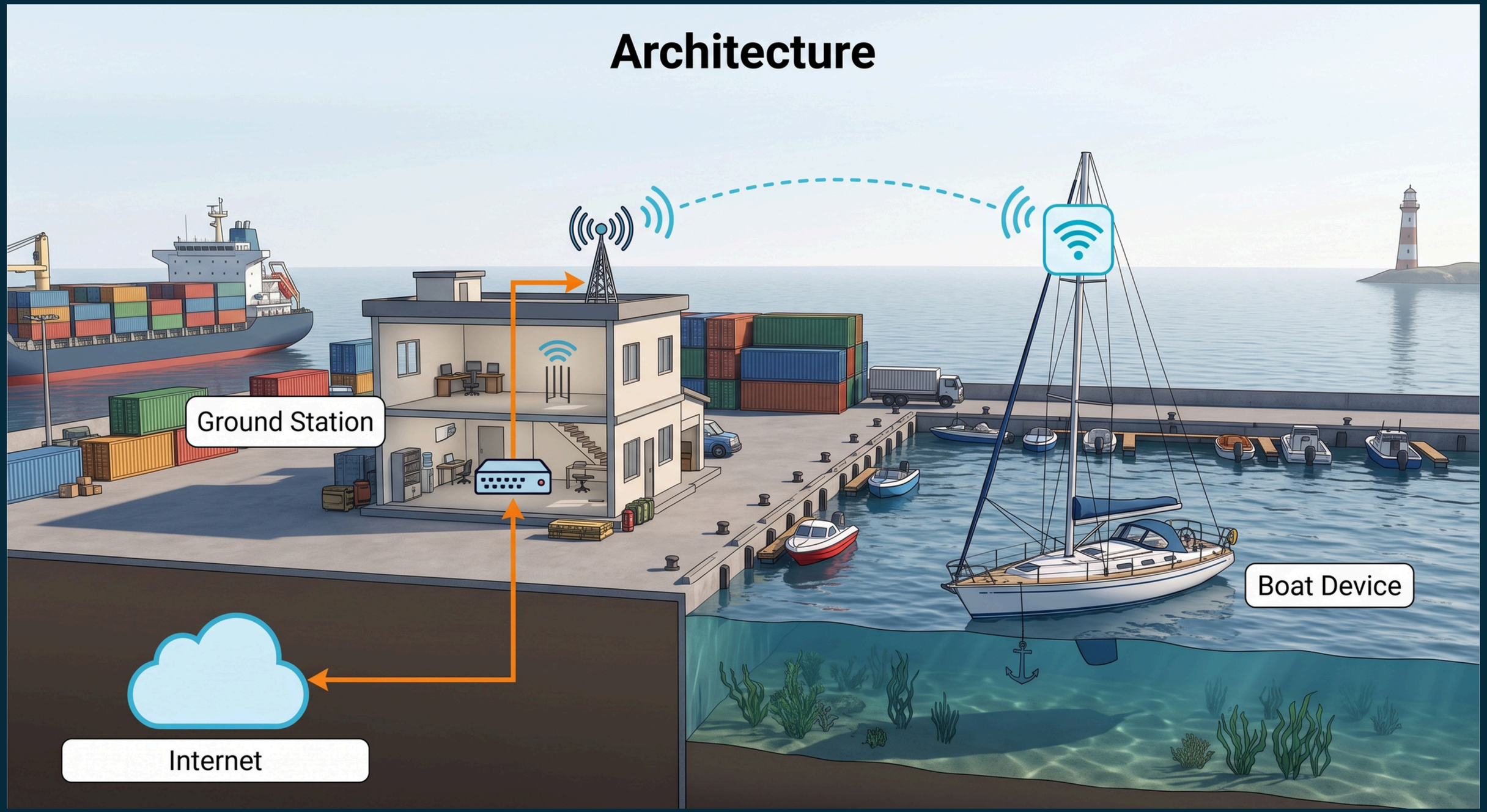
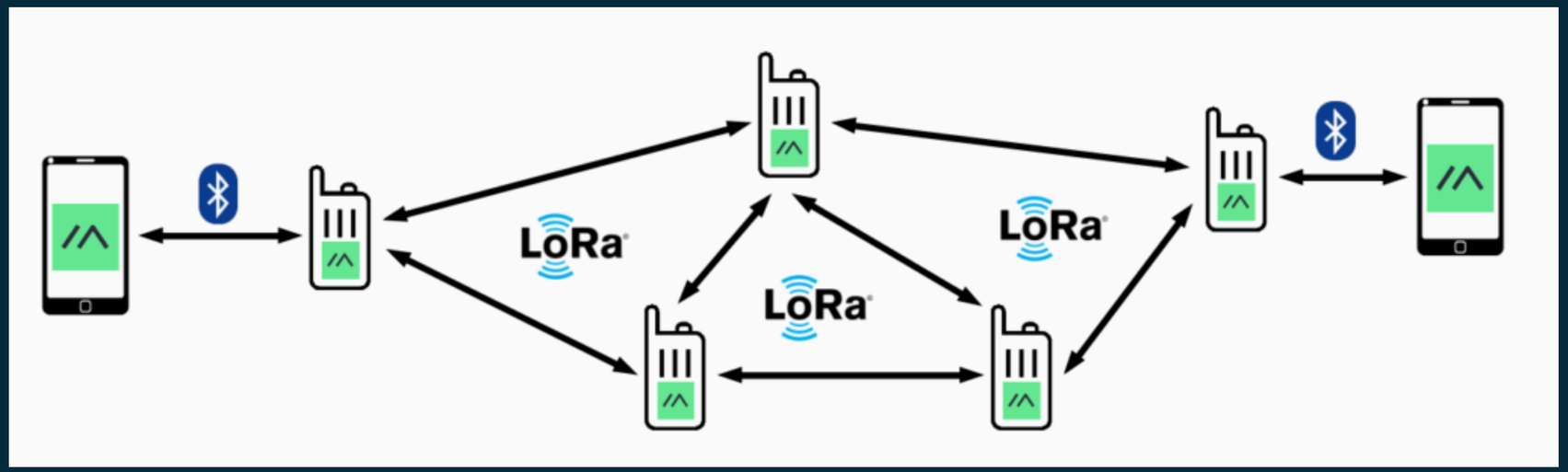
MeshBoat
Secure your floating home

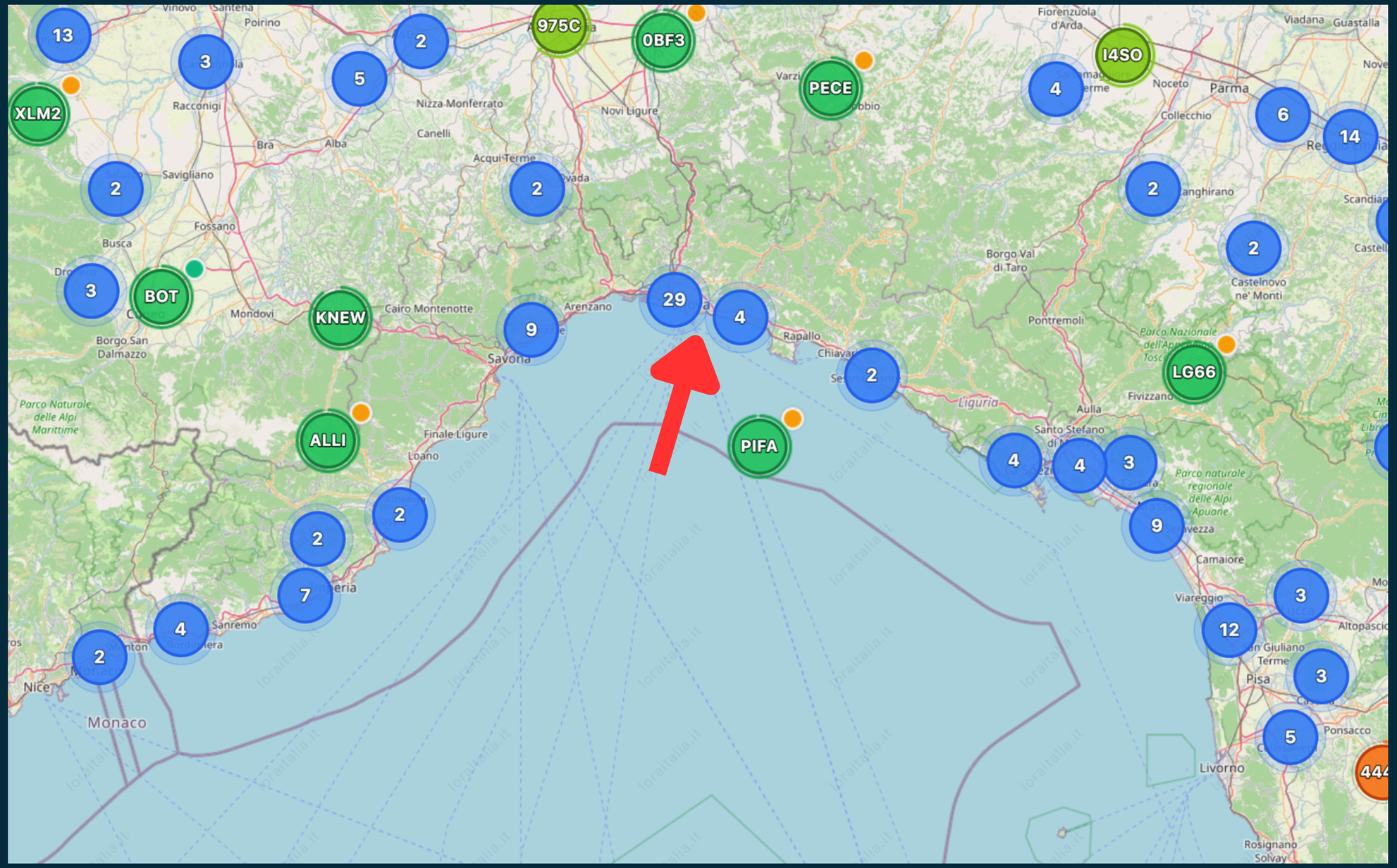
Proposed Solution

Architecture

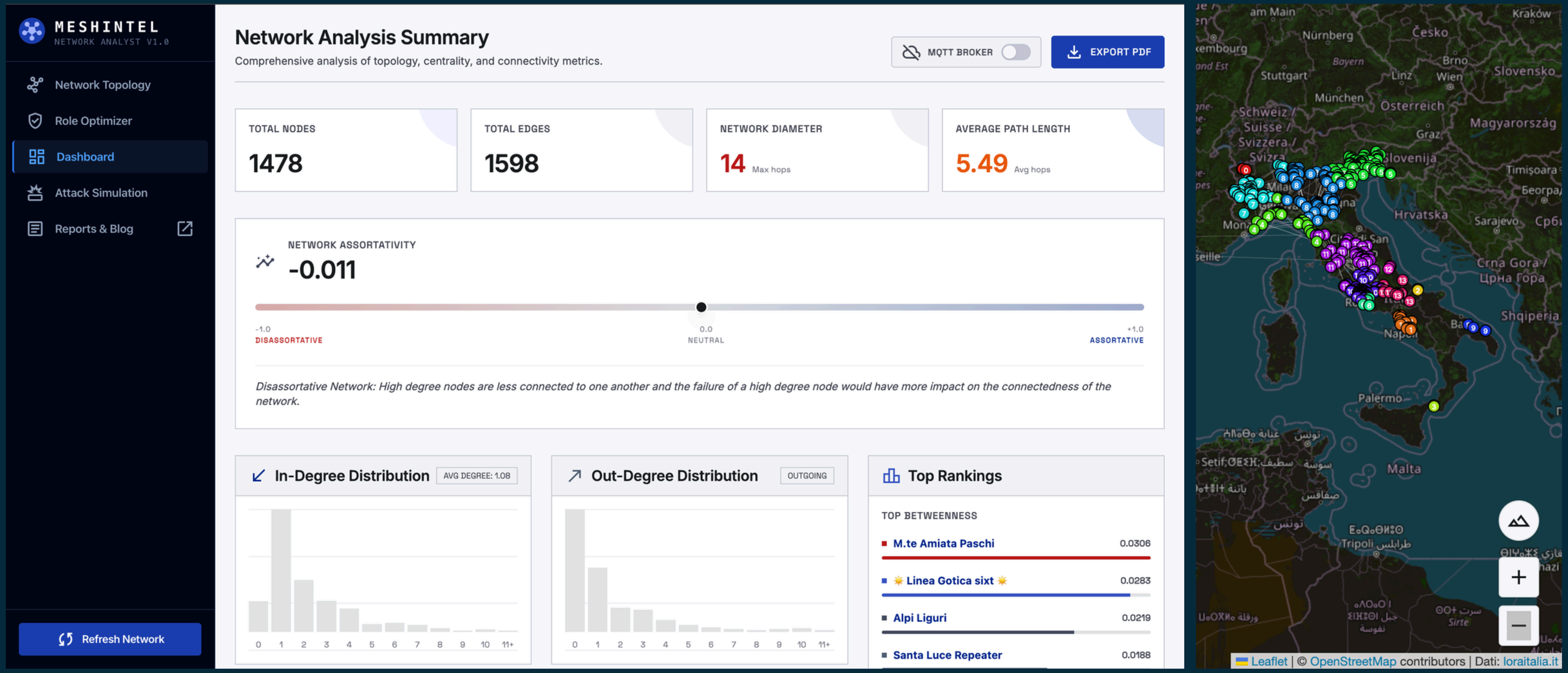




meshtastic.org



loraitalia.it



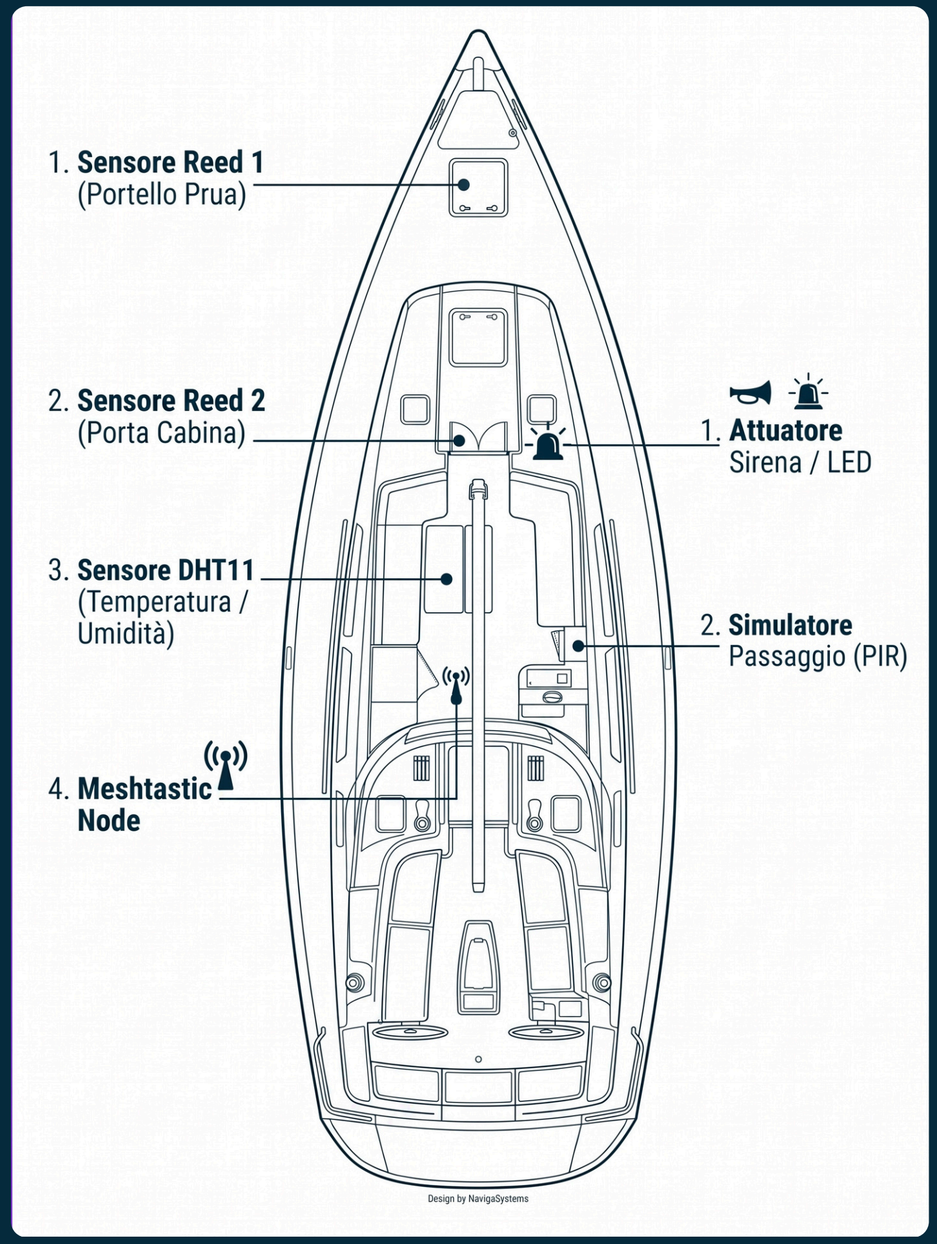
(our) Italian mesh network analysis - meshintel.it

Our Vision

A network of boats

Meshtastic in nautic

By connecting boats through a LoRa mesh network, we eliminate the need of traditional cellular coverage. This ensures that critical security alerts are always transmitted, fostering a safer and more connected nautical community.



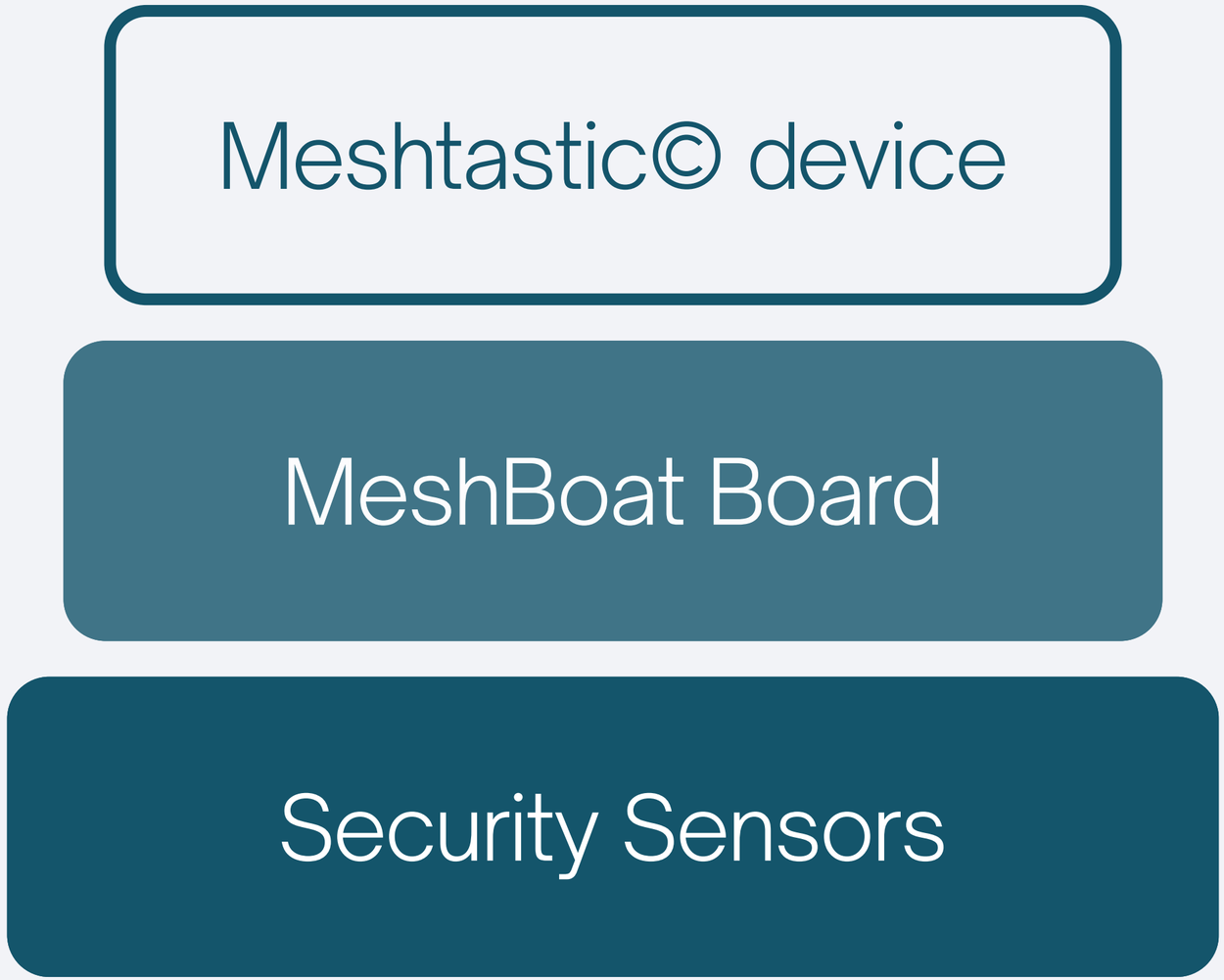
Security

- **Comprehensive Sensor Array:** Magnetic Reed switches (doors/hatches) and motion sensor that secure both perimeter and interior.
- **Environmental Monitoring:** A DHT11 sensor to track temperature and humidity levels.
- **Instant Local Deterrent:** A high-intensity siren and LED trigger immediately upon intrusion.
- **Real-Time LoRa Alerts:** The onboard Meshtastic node broadcasts alerts instantly, working without cellular coverage.

Architecture

Boat Device

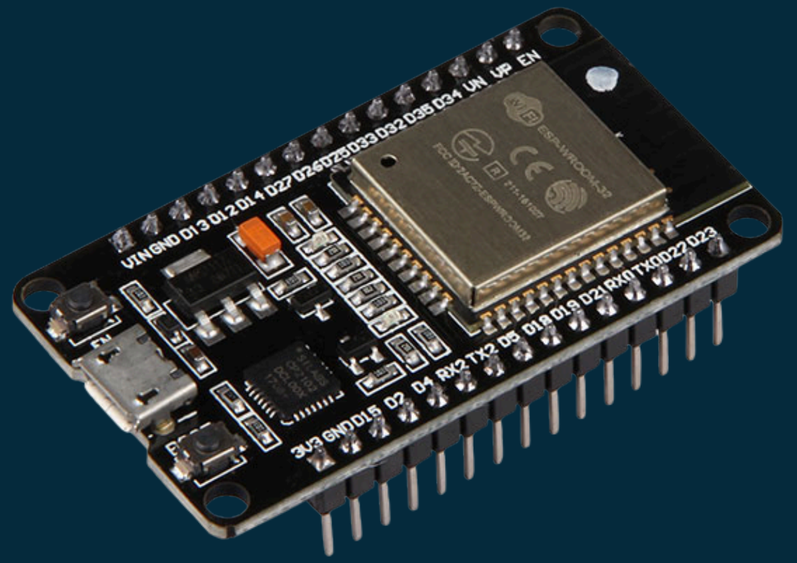
AI generated



solar powered

Meshtastic© device

MeshBoat Board



XIAO ESP32 powered

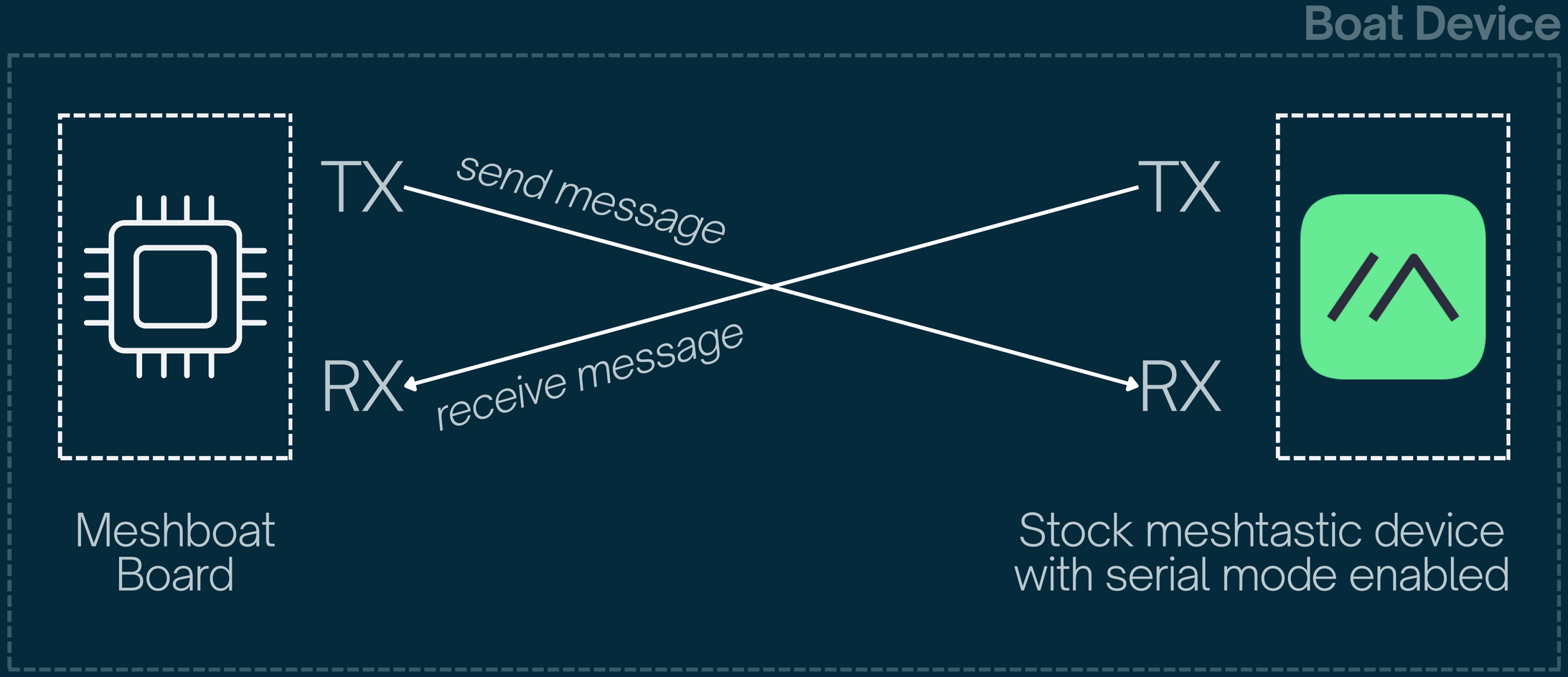
ESP32-wroom-32 powered

Why two MCUs?

Customizing Meshtastic firmware is a really **bad idea** !

- Breaking the mesh
- Isolation
- Updates
- Merging code
- Testing
- Mobile app
- ...

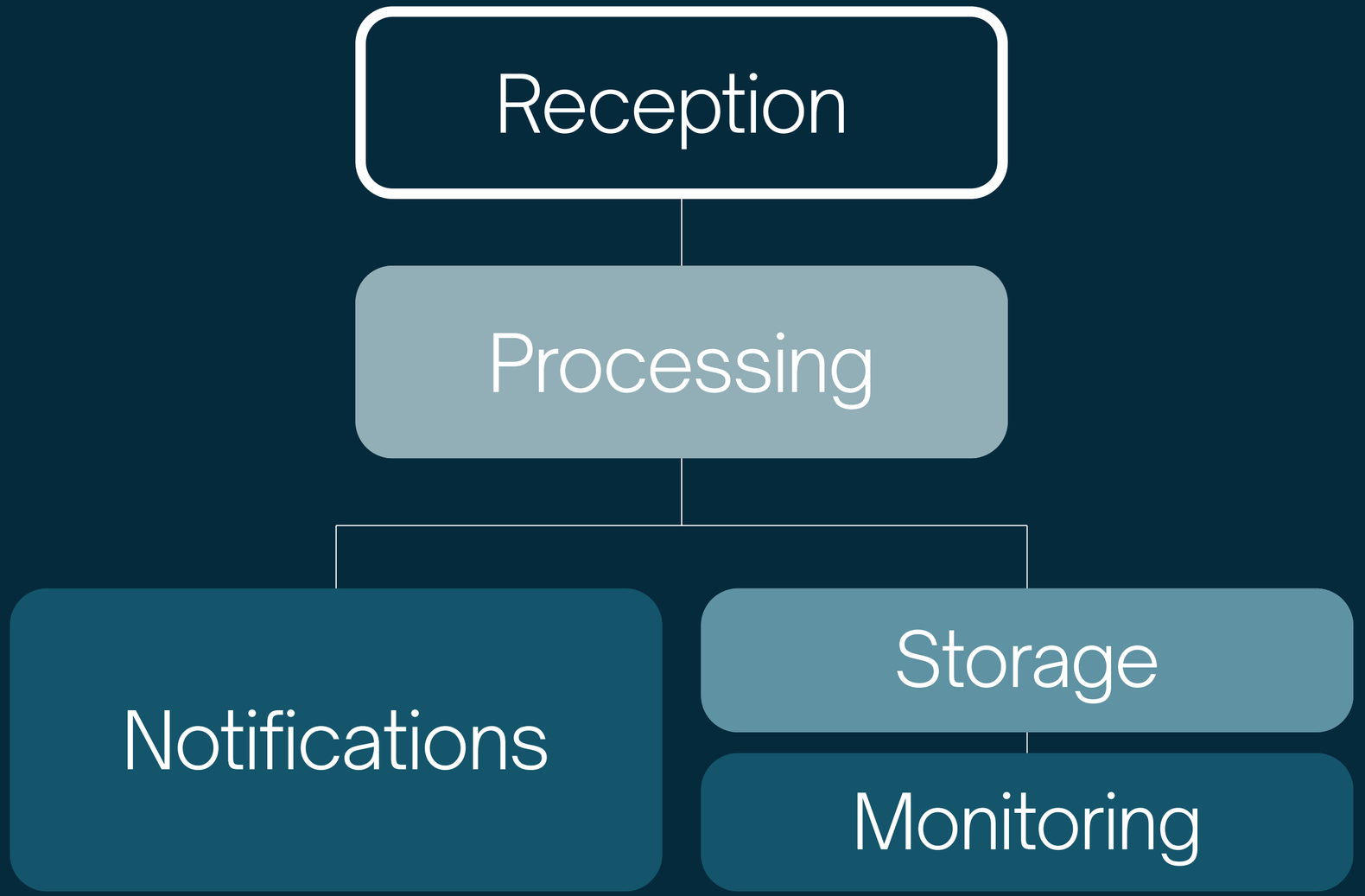
Just use UART !



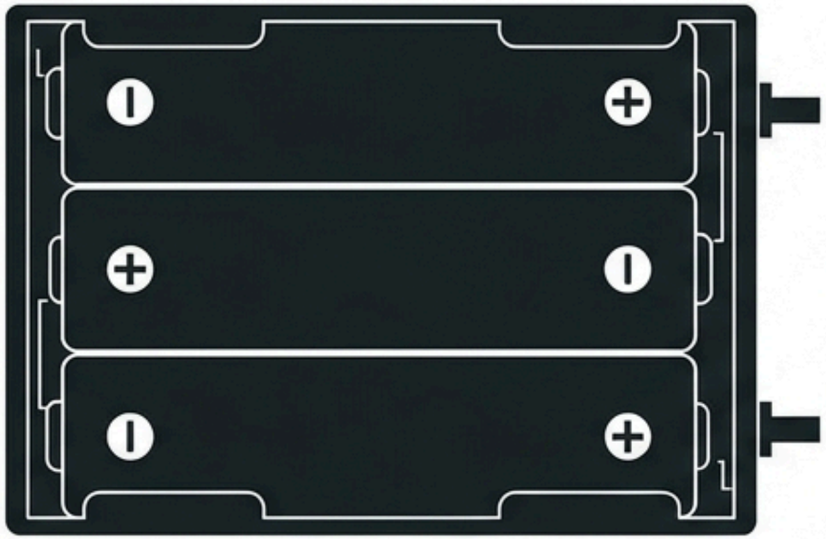
*All messages are sent on a private channel to avoid spam

Ground Station

AI generated

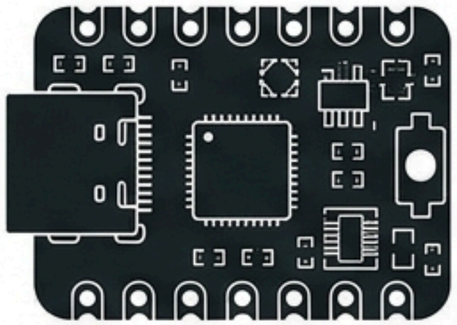

Reception
Processing
Notifications
Storage
Monitoring

Hardware Details & Tinkering

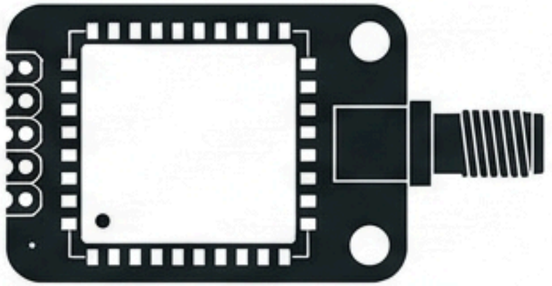


18650
pack

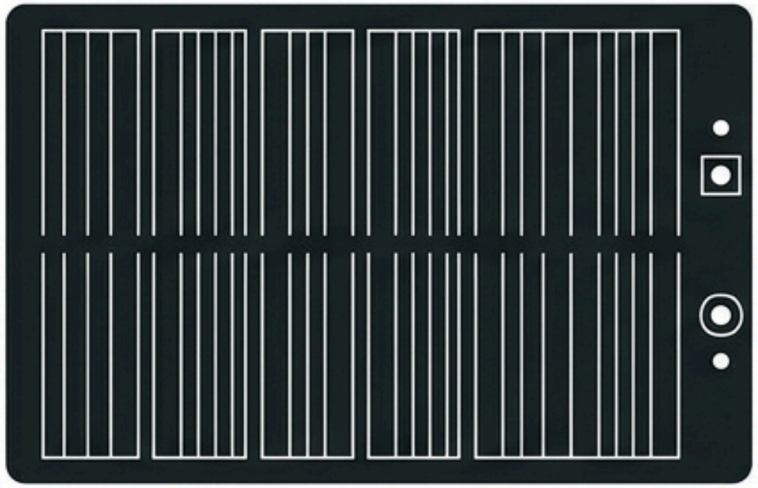
18650



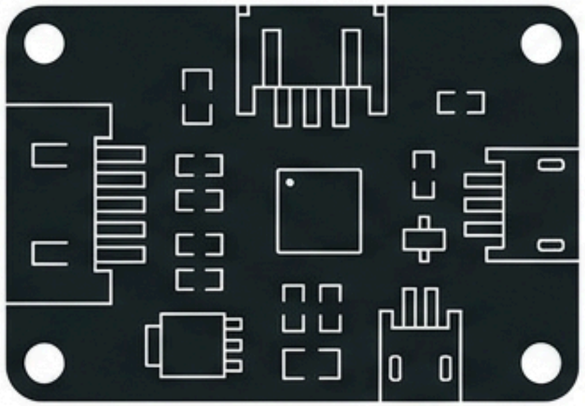
XIAO ESP32S3



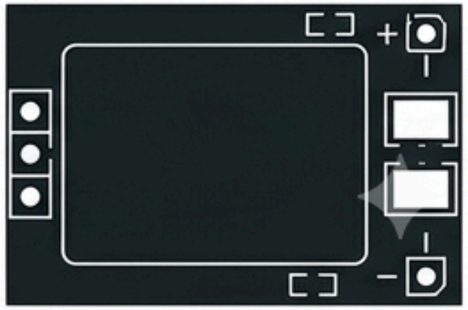
Wio-SX1262



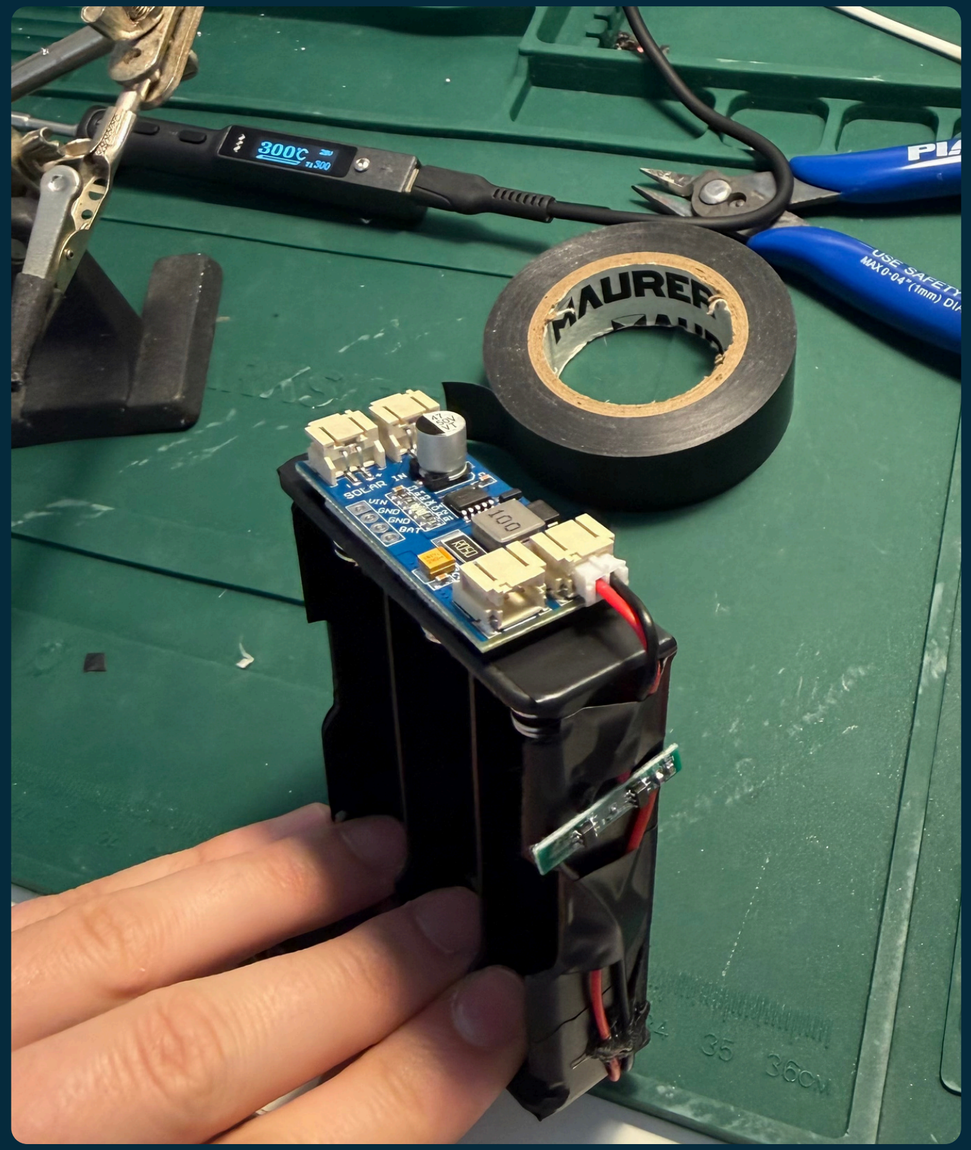
5W 6V solar panel



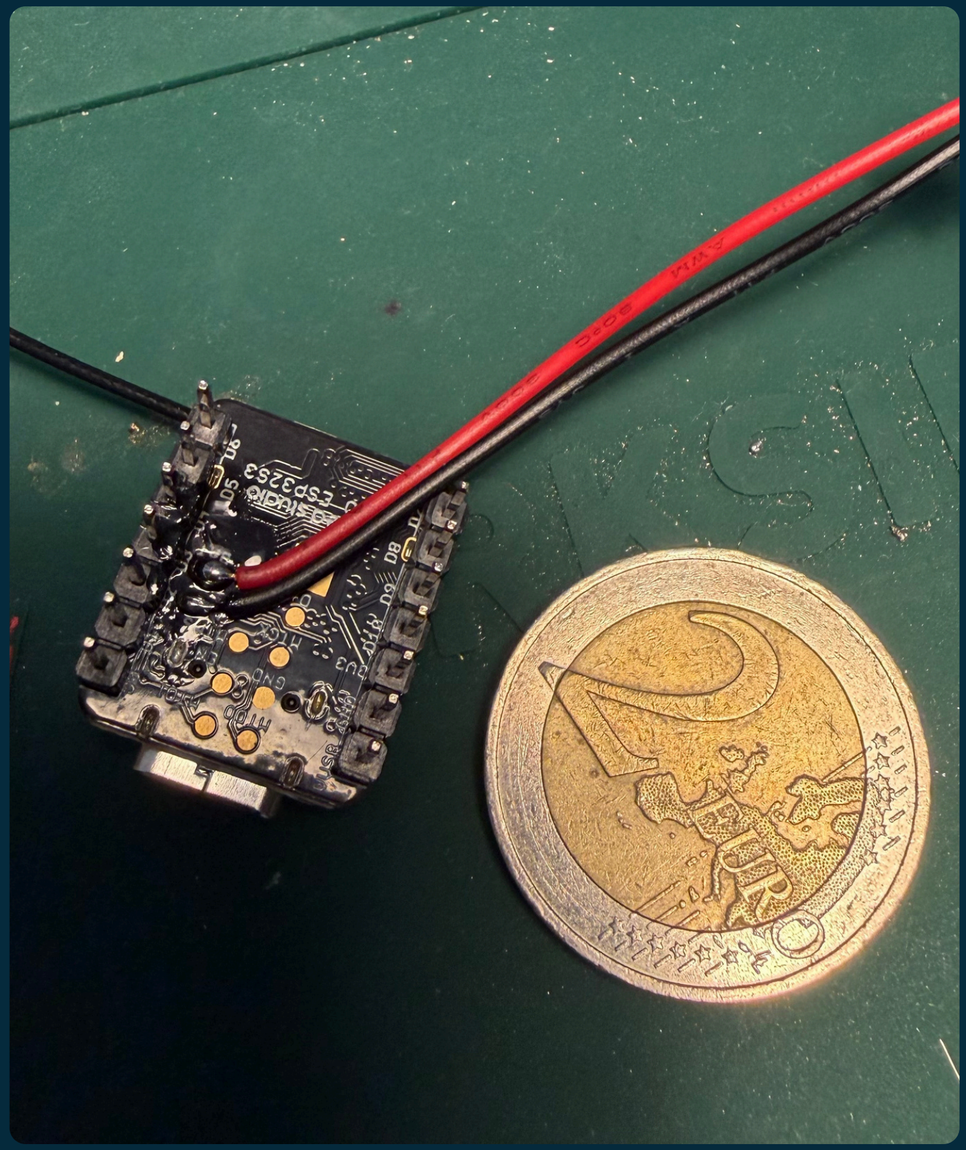
CN3791 solar charging board



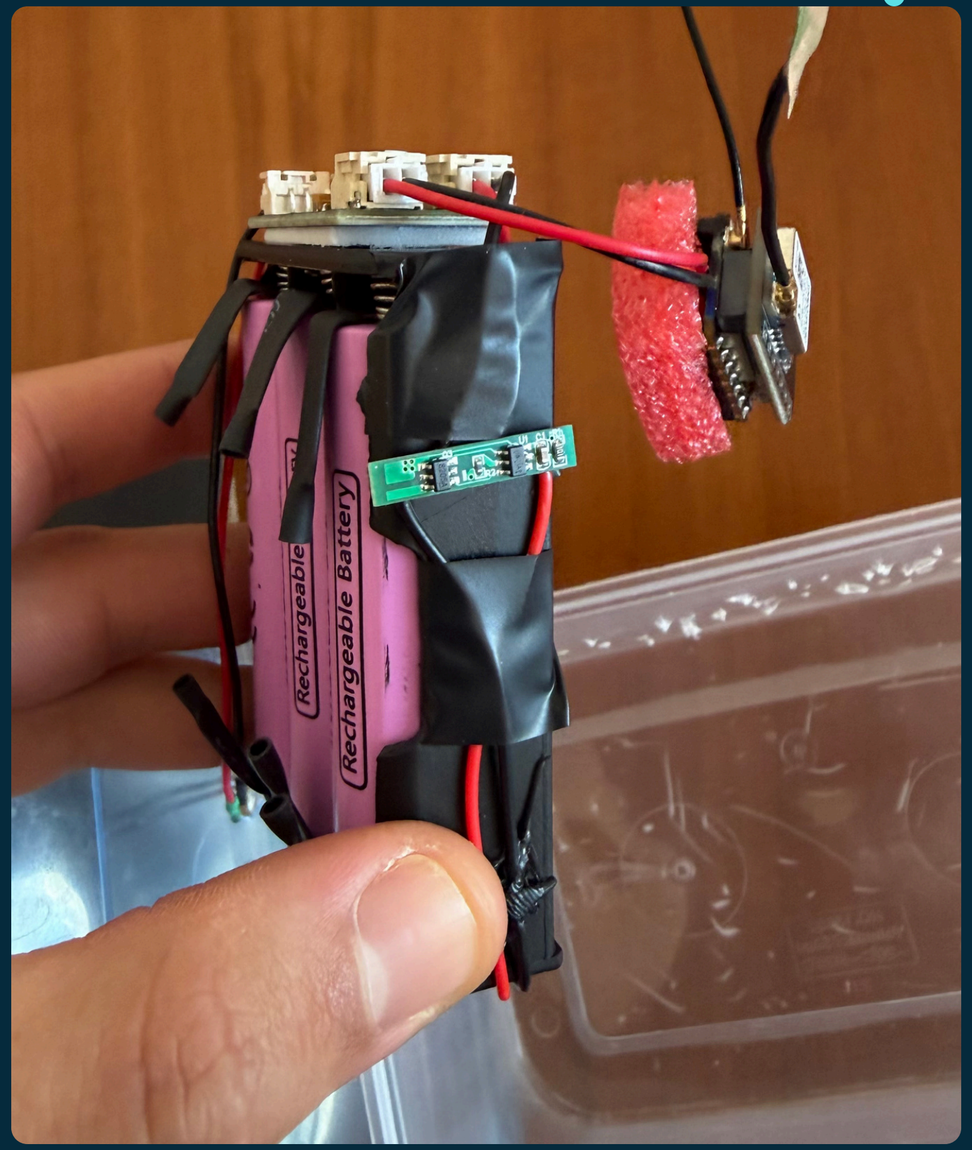
Battery BMS



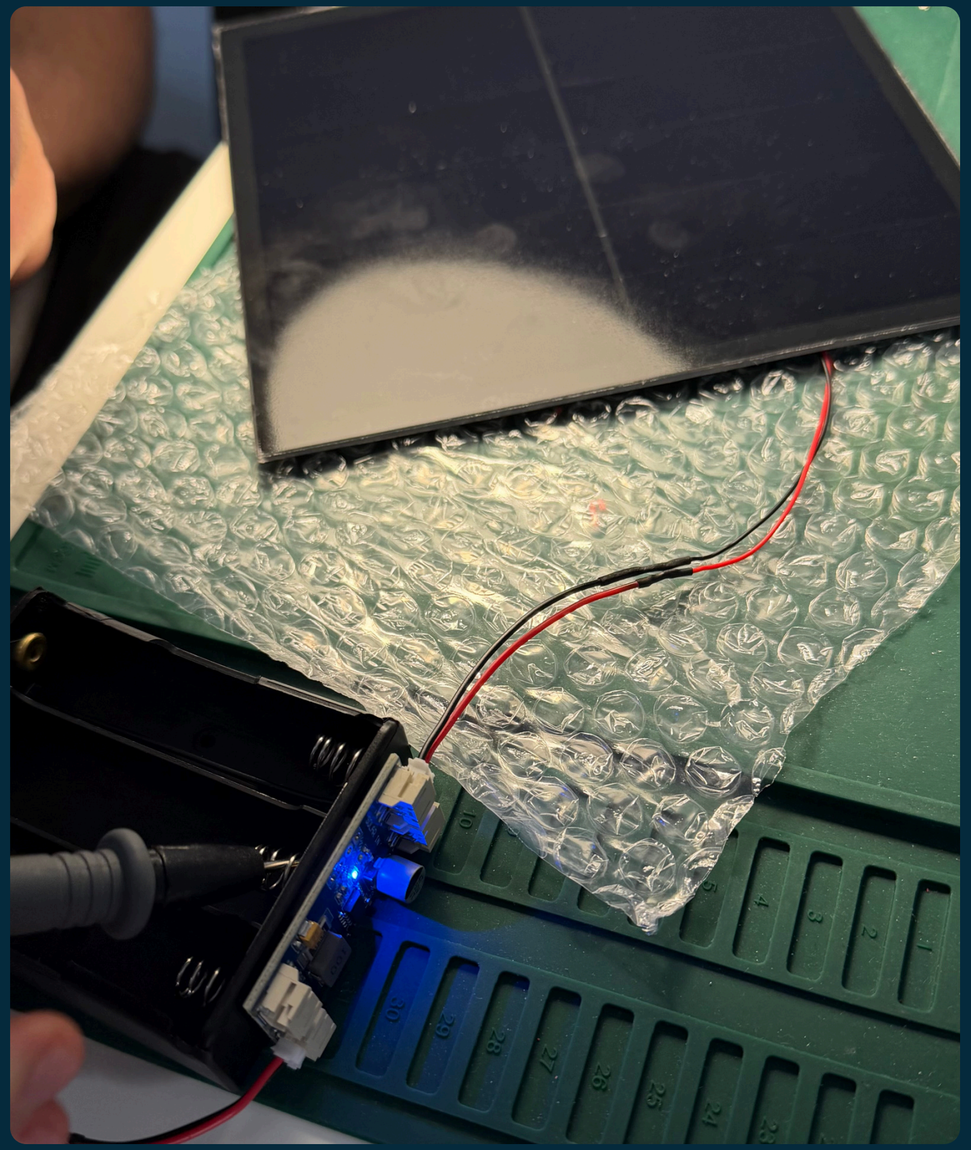
3x18650 battery pack with protection circuit



Soldering XIAO ESP32 battery pads

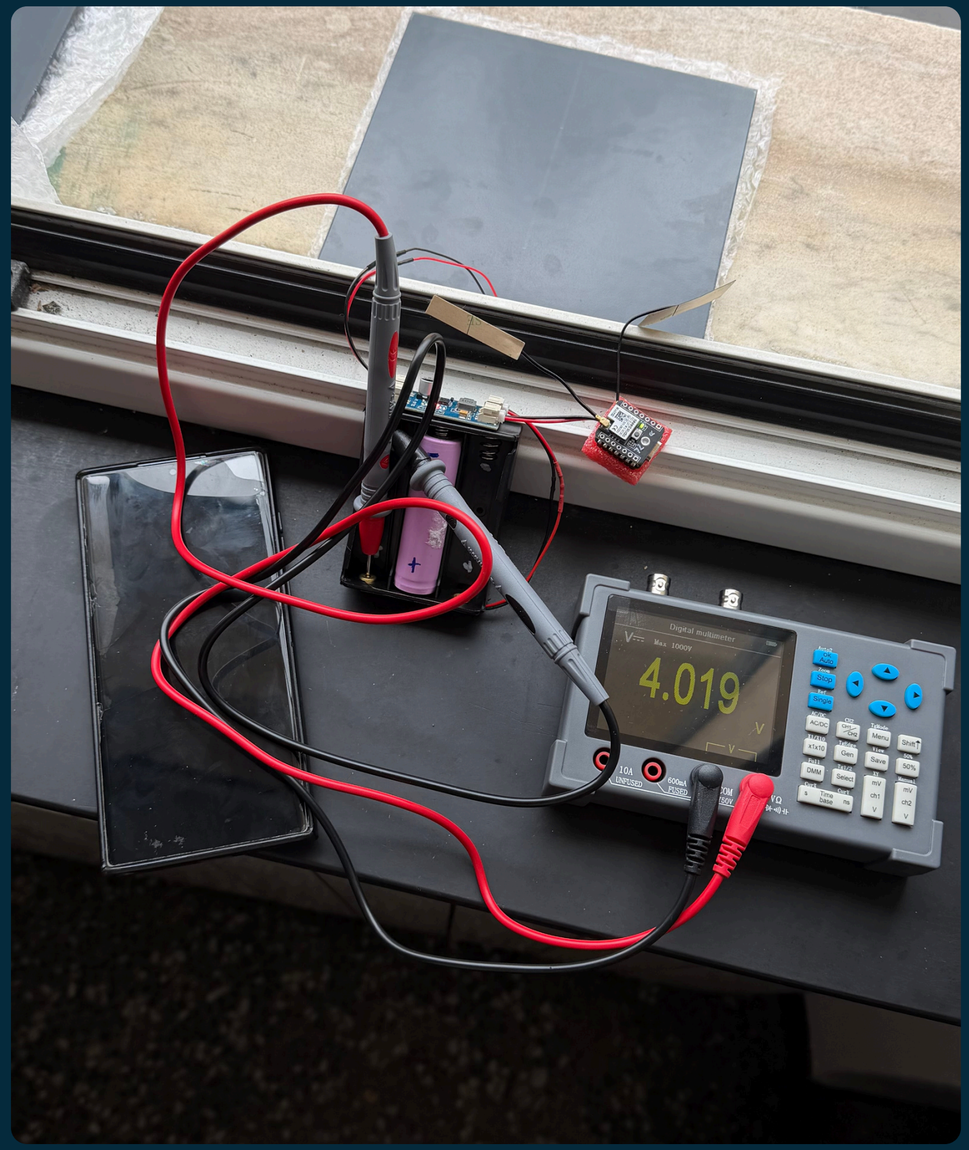


Battery powered node assembled

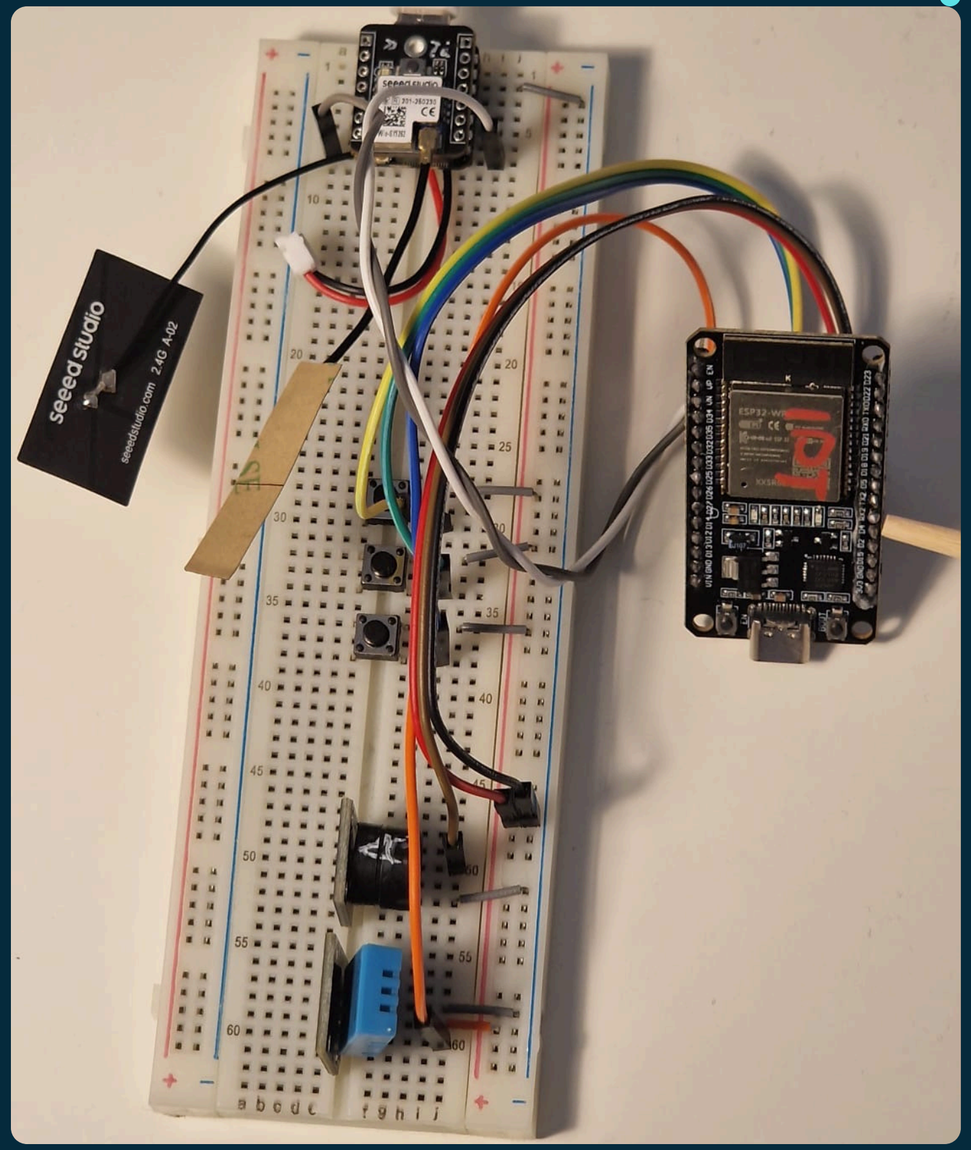


6V Solar Panel with (fake) MPPT circuit

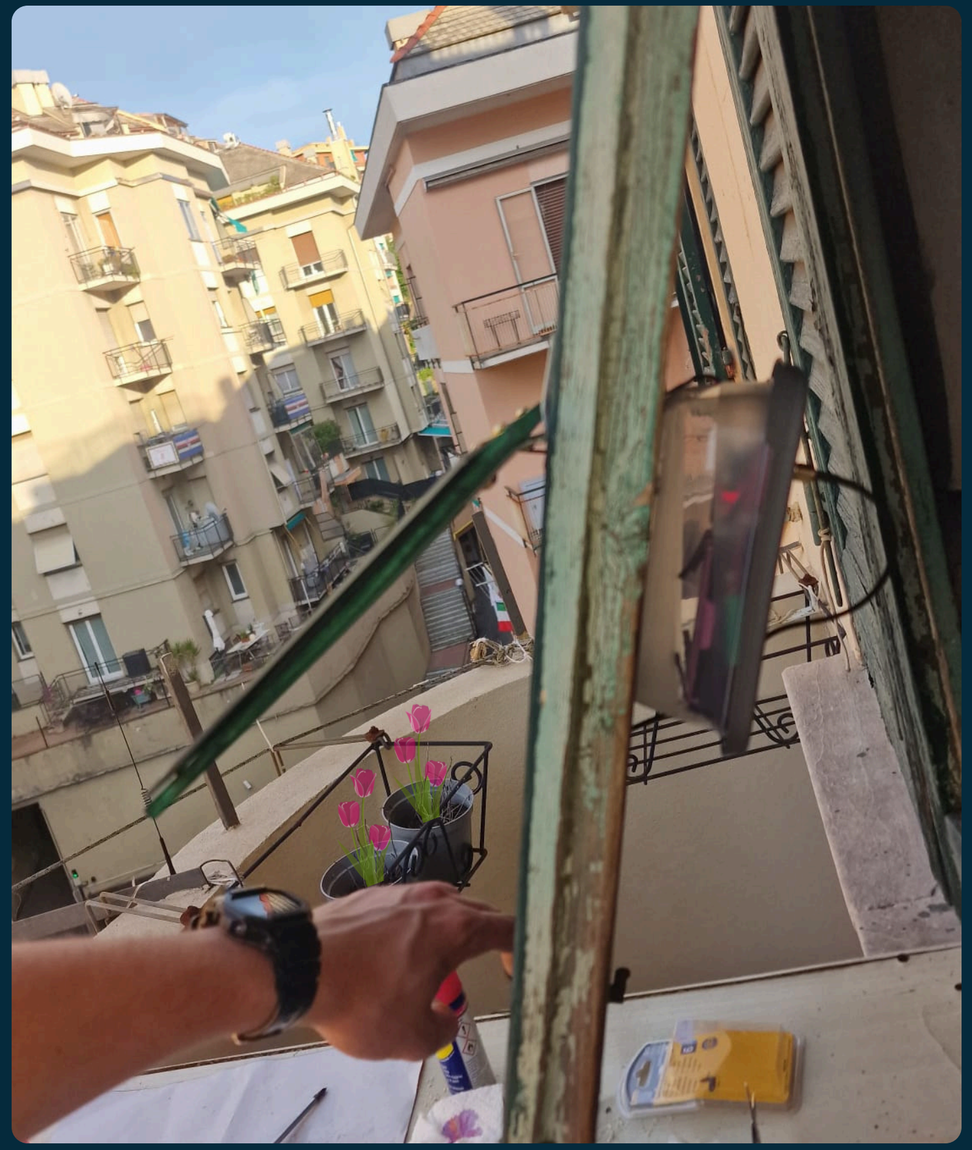
Learning a new skill



Succesfully charging a battery using the sun

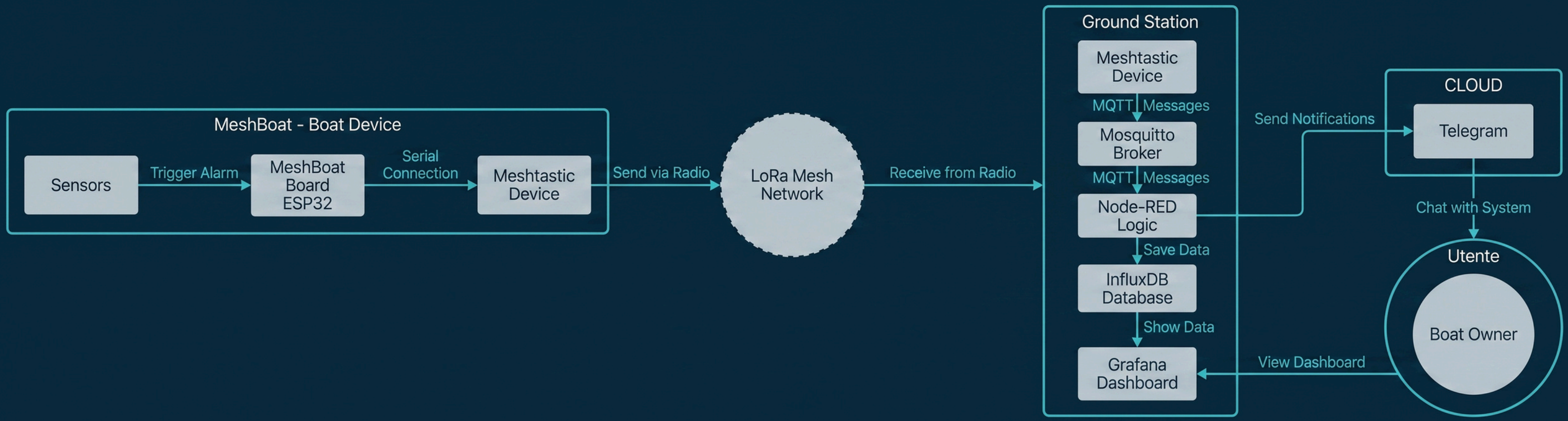


MeshBoat Board prototype



Deployed “Ground Station”

Hardware and Software togeter...



Software

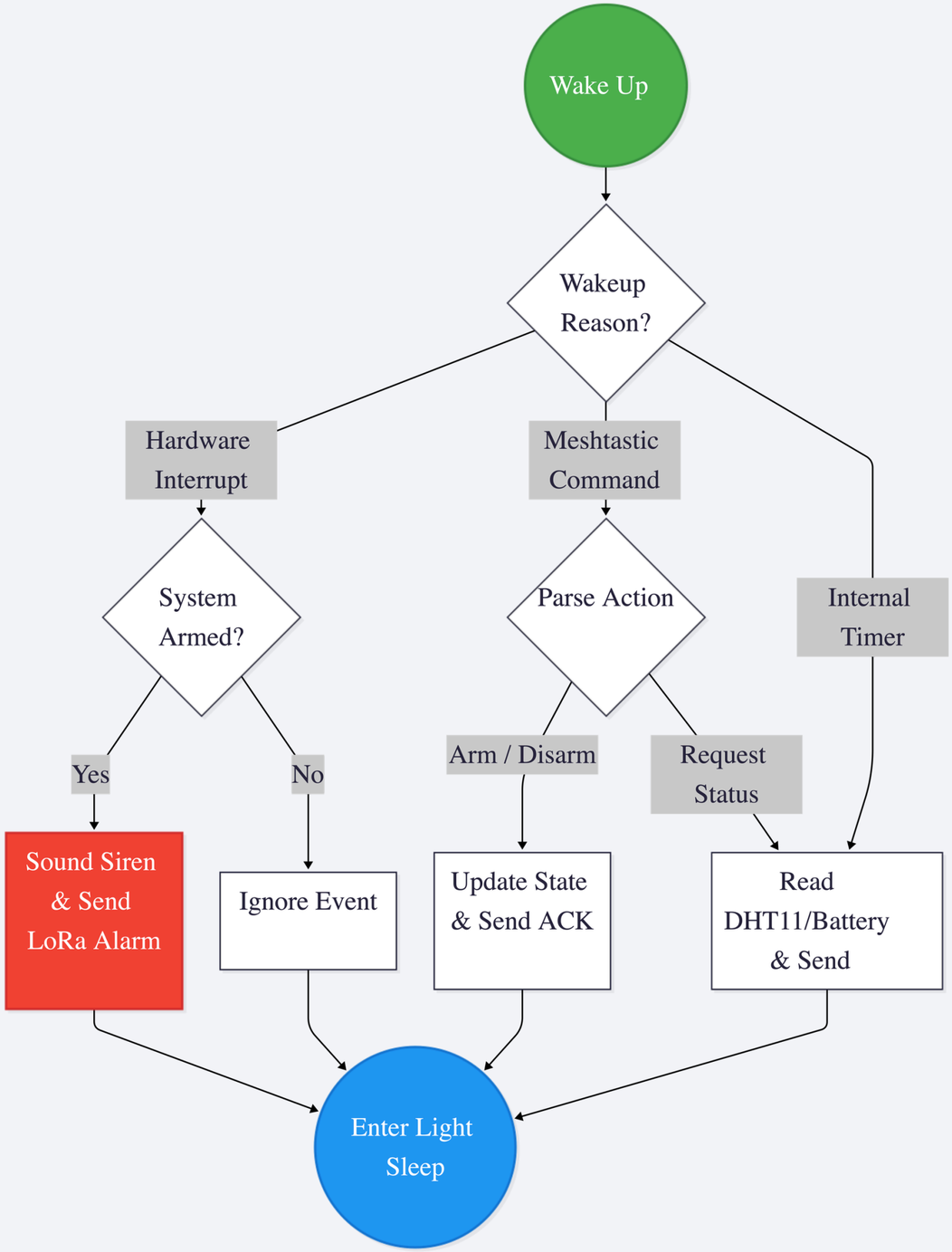
Pull-up
Pull-down

Reed door/hatch sensors & PIR motion sensor rest at 0, trigger to 1.
The Meshtastic radio's UART line rests at 1 (serial idle standard), drops to 0 when a message starts.

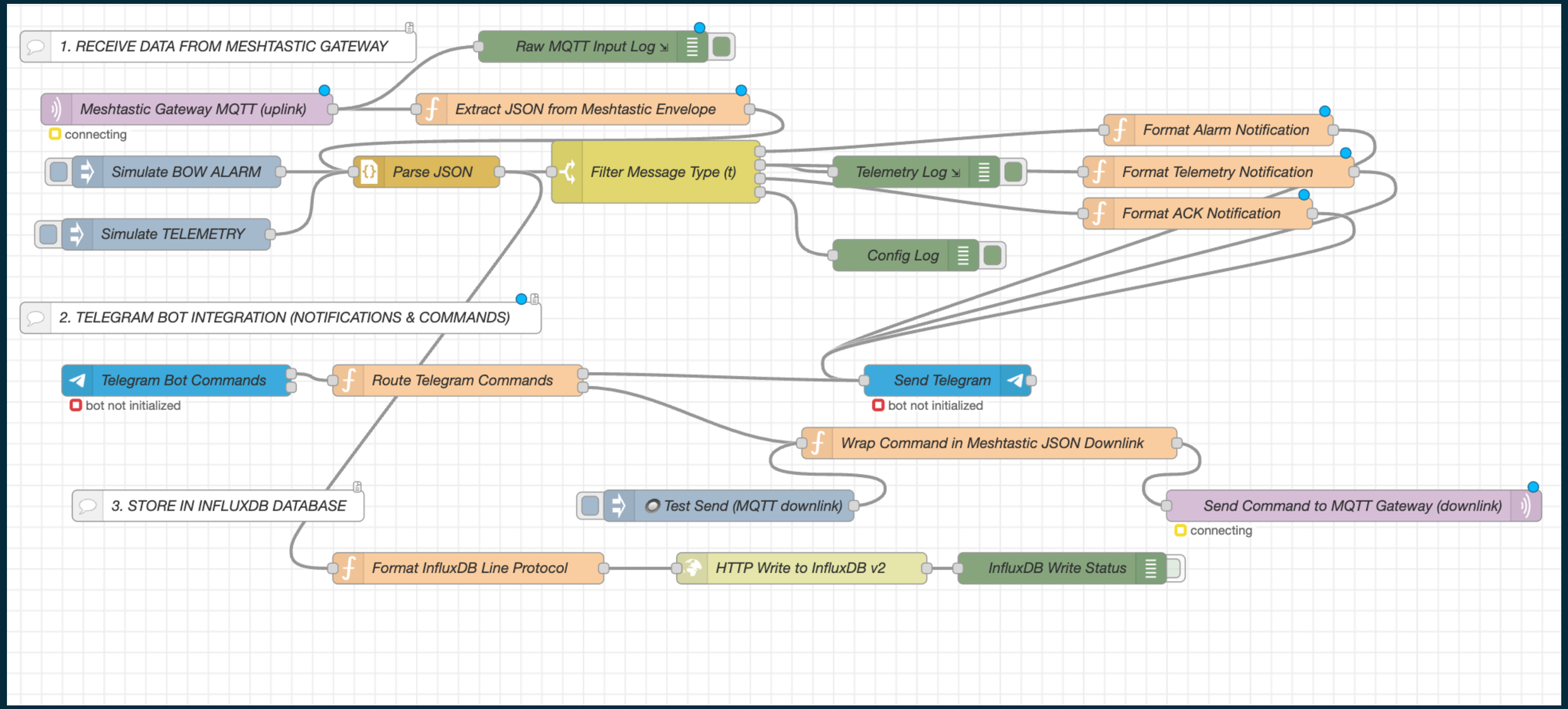
Interrupt

EXT1 wakes the ESP32 when a physical sensor triggers (hatch, cabin door, PIR).
EXT0 wakes it when a remote command arrives over radio (UART RX goes LOW).

Flow Chart


Wake Up
Wakeup Reason?
Hardware Interrupt
Meshtastic Command
Internal Timer
System Armed?
Parse Action
Yes
No
Arm / Disarm
Request Status
Sound Siren & Send LoRa Alarm
Ignore Event
Update State & Send ACK
Read DHT11/Battery & Send
Enter Light Sleep

Nodered Flow



Visualization & Interaction

MeshBoat DASHBOARD

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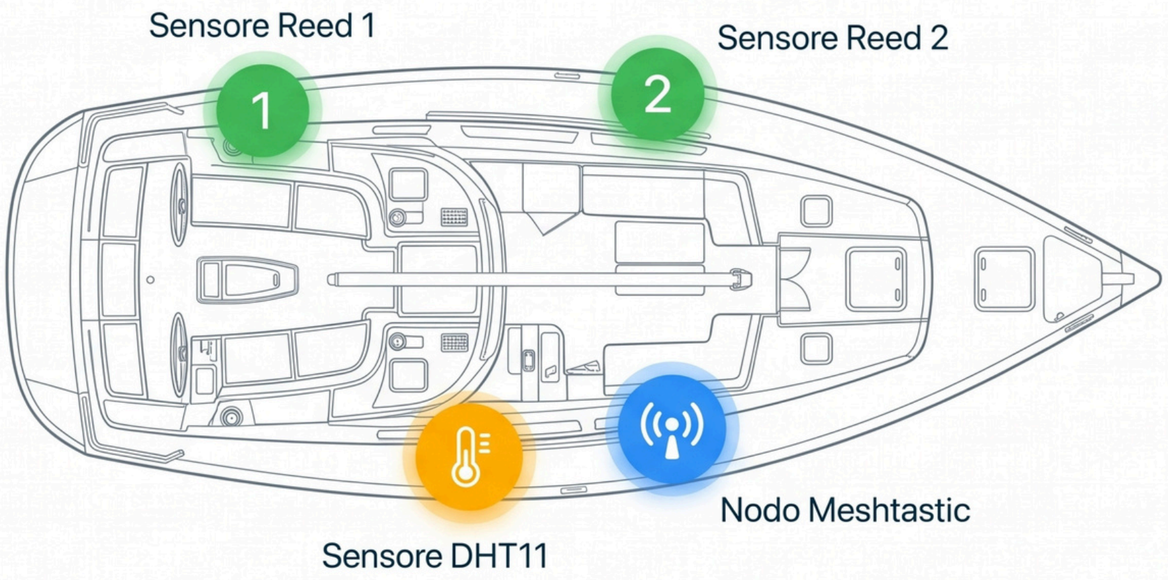


STATO ON-LINE

SICURO

AMBIENTE

22 °C 54 %

RETE LoRa

CONNESSO

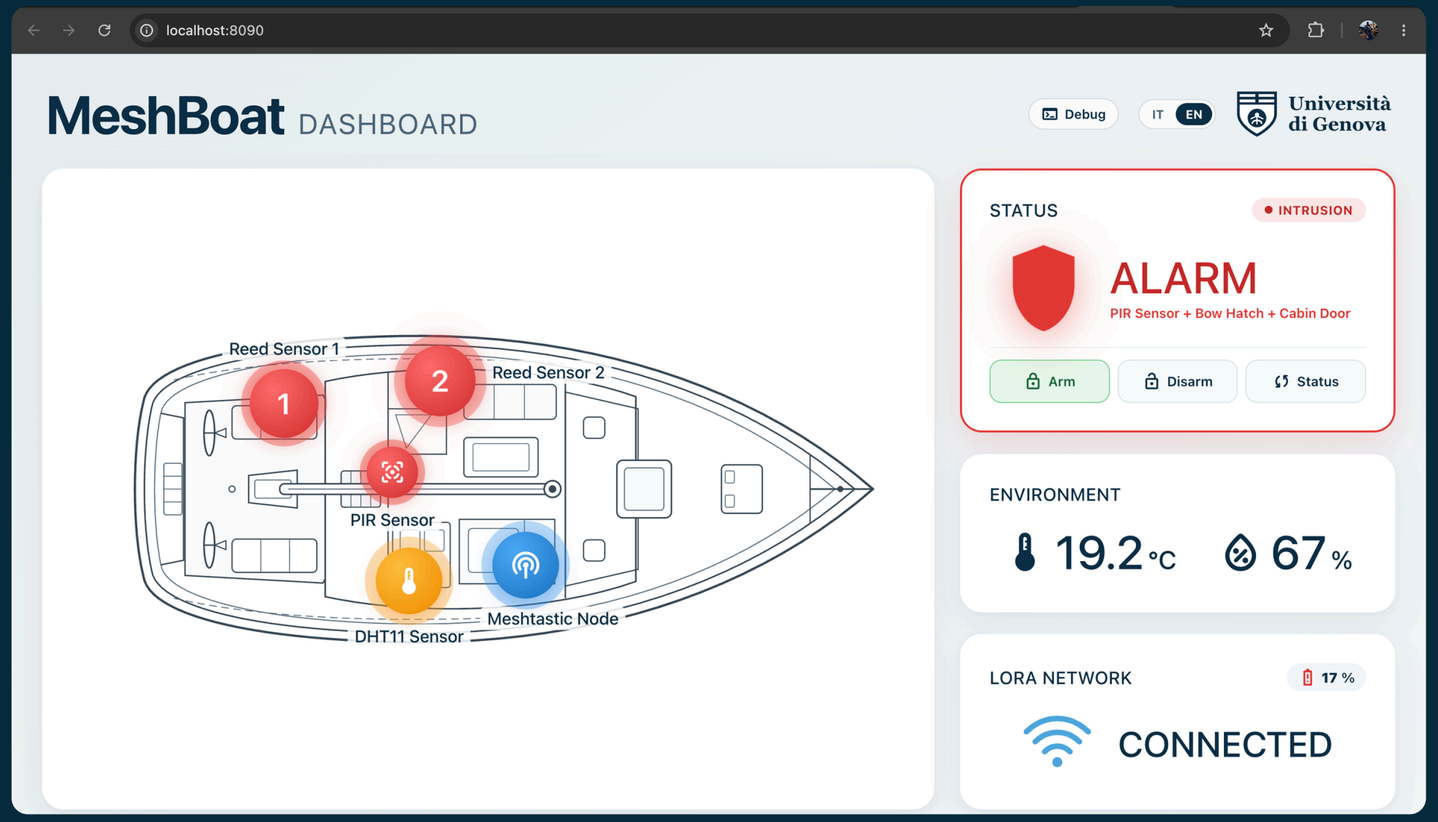
demo ↗

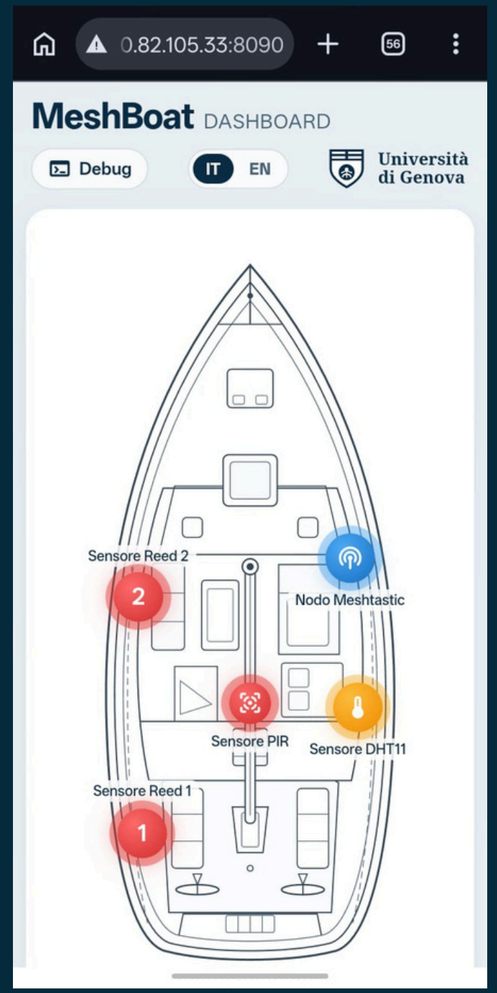
Telegram Bot

Stay Connected, Anywhere

Control your boat's security system directly from your smartphone using our dedicated Telegram Bot.
Available Commands:

- **/arm** – Instantly activates the onboard security perimeter and sensors.
- **/disarm** – Deactivates the alarm system to allow safe entry.
- **/status** – Requests real-time telemetry (temperature, humidity, battery) and the current system state.





Responsive Design

Conclusion

Improvements & Scalability

MeshBoatGuard is a low-budget prototype demonstrating the viability of a LoRa mesh network for maritime security, with a fully scalable architecture. Future upgrades:

- **Real sensors** instead of prototype push-buttons
- **Real battery telemetry**
- **High-gain marine antenna** for extended range
- **More powerful solar panel** for 24/7 autonomy even in cloudy weather

Thank You !